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Modular Unit with Camera and Display

Abstract

A device or unit with a display and camera is provided. The device includes coupling components that couple the device to stackable storage unit containers. In one embodiment, the device includes a battery interface to couple to a battery for power tools, a camera, and a display that displays images captured by the camera.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION [0001] The present application claims the benefit of and priority to U.S. Provisional Application 63/559,366 filed on Feb. 29, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The present disclosure is directed generally to the field of tool storage systems and related devices. The present disclosure relates specifically to a device or unit that includes a camera, a display, and a coupling mechanism to detachably couple the device to another device or container, such as a tool storage container, in a modular tool storage system.

[0003] Tool storage units are often used to transport tools and tool accessories. Some storage units are designed to incorporate into a modular storage system. Within a modular storage system, different units, devices and/or containers may provide varying functions, such as being adapted to capture images via a camera and display the images on a display.

SUMMARY OF THE INVENTION

[0004] One embodiment of the invention relates to a utility module including a housing, a bottom surface defined by the housing, a plurality of male couplers extending from the bottom surface, the plurality of male couplers detachably and rigidly coupling the housing to a top surface of a storage unit, a display pivotally coupled to the housing, a cable configured to extend and retract from within the housing, and a camera coupled to an outer end of the cable. The display displays one or more images captured by the camera.

[0005] Another embodiment of the invention relates to a utility module including a housing, a bottom surface defined by the housing, a plurality of male couplers extending from the bottom surface, the plurality of male couplers detachably and rigidly coupling the housing to a top surface of a storage unit, a display pivotally coupled to the housing, a camera, the display displaying one or more images captured by the camera, and a battery interface configured to physically couple to a rechargeable power tool battery. The display and camera are configured to receive power from the rechargeable power tool battery coupled to the battery interface.

[0006] Another embodiment of the invention relates to a utility module including a housing, a bottom surface defined by the housing, a plurality of male couplers extending from the bottom surface, the plurality of male couplers detachably and rigidly coupling the housing to a top surface of a storage unit, a display pivotally coupled to the housing, a storage area defined by the housing, the storage area configured to store equipment, and a camera. The display displays one or more images captured by the camera.

[0007] Another embodiment of the invention relates to a device including a housing, a bottom surface defined by the housing, a plurality of male couplers extending from the bottom surface, a display pivotally coupled to the housing, and a camera operatively coupled to the display such that images captured by the camera are viewable on the display. In a specific embodiment, the device includes a cable configured to extend and retract from within the housing, and a camera coupled to the cable. The plurality of male couplers detachably and rigidly couple the housing to a top surface of a storage unit. The display includes a screen that displays one or more images captured by the camera.

[0008] In various embodiments, the device includes a first lid pivotally coupled to the housing, the first lid and the housing collectively defining a first compartment. The cable extends out of and retracts into the first compartment.

[0009] In various embodiments, the display is pivotally coupled to the housing via the display being pivotally coupled to the first lid and the first lid being pivotally coupled to the housing.

[0010] In various embodiments, the device includes a second lid pivotally coupled to the housing, the second lid and the housing collectively defining a second compartment, and the device further includes a battery interface in the second compartment configured to physically couple to a rechargeable power tool battery. The display and camera are configured to receive power from the

rechargeable power tool battery.

[0011] In various embodiments, each of the plurality of male couplers include a body extending below the bottom surface, a first tongue, and a second tongue. The first tongue and the second tongue both extend from the body and are both offset from and below the bottom surface. The first tongue defines a first channel between the bottom surface and the first tongue, and the second tongue defines a second channel between the bottom surface and the second tongue. The first channel and second channel each extend on opposing sides of the body, and each channel includes a front open end and a back closed end.

[0012] Another embodiment of the invention relates to a device including a housing, a bottom surface defined by the housing, a plurality of male couplers extending from the bottom surface, a display pivotally coupled to the housing, a camera operatively coupled to the display such that images captured by the camera are viewable on the display, and a battery interface configured to physically couple to a rechargeable power tool battery. The display includes a screen that displays one or more images captured by the camera. The plurality of male couplers detachably and rigidly couple the housing to a top surface of a storage unit. The display and camera are configured to receive power from the rechargeable power tool battery coupled to the battery interface.

[0013] Additional features and advantages will be set forth in the detailed description which follows, and will be readily apparent to those skilled in the art from the description or recognized by practicing the embodiments as described in the written description and/or shown in the accompany drawings. It is to be understood that both the foregoing general description and the following detailed description are exemplary.

[0014] The accompanying drawings are included to provide further understanding and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiments and, together with the description, serve to explain principles and operation of the various embodiments. In addition, alternative exemplary embodiments relate to other features and combinations of features as may be generally recited in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] This application will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements in which:

[0016] FIG. 1 is a perspective view of a device including a display and camera, according to an exemplary embodiment.

[0017] FIG. 2 is a perspective view of the device of FIG. 1, according to an exemplary embodiment.

[0018] FIG. 3 is a perspective view of the device of FIG. 1, according to an exemplary embodiment.

[0019] FIG. 4 is a perspective view of the device of FIG. 1, according to an exemplary embodiment.

[0020] FIG. 5 is a perspective view from below of the device of FIG. 1, according to an exemplary embodiment.

[0021] FIG. 6 is a detailed perspective view from below of the device of FIG. 1, according to an exemplary embodiment.

[0022] FIG. 7 is a front perspective view of the device of FIG. 1 with the first and second lids removed, according to an exemplary embodiment.

[0023] FIG. 8 is a rear perspective view of the device of FIG. 1 with the first and second lids removed, according to an exemplary embodiment.

[0024] FIG. **9** is a front view of a portion of graphics and images shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

[0025] FIG. **10** is a front view of a portion of graphics and images shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

[0026] FIG. **11** is a front view of a portion of graphics and images shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

[0027] FIG. **12** is a front view of a portion of graphics shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

[0028] FIG. **13** is a front view of a portion of graphics and images shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

[0029] FIG. **14** is a front view of a portion of graphics and images shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

[0030] FIG. **15** is a front view of a portion of graphics and images shown on the display of the device of FIG. **1**, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0031] Referring generally to the figures, an embodiment of a stackable device, container or unit is shown. One or more of the devices are configured to selectively couple and decouple with storage units. The stackable device couples with storage units and includes a camera and a display to show images captures by the camera. The display is pivotal with respect to the housing and includes an interface to receive power from a power tool battery. Applicant has observed that providing a single device that includes a camera, a display that displays images from the camera, and modular coupling components facilitates users transporting the device with other units in a modular system and using the camera to view objects that would otherwise be difficult to observe.

[0032] Referring to FIGS. **1-4**, various aspects of utility module, shown as device **110**, are shown. Device **110** includes housing **120**, first lid **122** pivotally coupled to housing **120**, and second lid **160** pivotally coupled to housing **120**. First toggle locking latch **126** secures first lid **122** in the closed position against housing **120**, and second toggle locking latch **162** secures second lid **160** in the closed position against housing **120**.

[0033] Device **110** includes display **130** coupled to housing **120**, such as pivotally coupled to housing **120**, such as via display **130** being pivotally coupled to first lid **122** and first lid **122** being pivotally coupled to housing **120**. Display **130** pivots with respect to housing about first axis **132** and second axis **134**. Device **110** includes cable **136** configured to extend and retract from housing **120** via cable cover **144** from an interior end coupled to housing **120** to an outer end **138**. Device **110** includes camera **140** coupled to cable **136**, such as to outer end **138** of cable **136**, the camera **140** receives light transmitted in the direction of outer end **138** of cable **136** (e.g., via camera **140** being at outer end **138** of cable, or via cable **136** being fiber optic and transmitting light to camera **140** at an interior end of cable). Arm **146** detachably engages with cable cover **144** to secure cable cover **144**. Cable **136** extends through cable cover **144** and is extended and retracted from housing **120** via cable cover **144**. Screen **180** of display **130** displays one or more images captured by camera **140**.

[0034] Display **130** pivots with respect to first lid **122** and housing **120** about first axis **132**, and display **130** pivots with respect to first lid **122** and housing **120** about second axis **134** distinct from first axis **132**. In various embodiments, second axis **134** is perpendicular to first axis **132**. In various embodiments, display **130** pivots 150 degrees with respect to housing **120** about first axis **132**, and display **130** pivots 180 degrees with respect to housing **120** about second axis **134**.

[0035] In various embodiments, display **130** includes a power button **184** configured to toggle whether screen **180** and camera **140** are powered. In various embodiments, display **130** is clipped and/or snapped against first lid **122**, such as when display **130** is in the retracted position (FIG. **1**). In various embodiments, display **130** includes a sealing gasket to provide water resistance and/or a watertight seal to protect against water entering display **130**. In various embodiments, display **130**

and screen **180** is a touchscreen device, thereby permitting users to touch screen **180** to provide input to device **110**.

[0036] Referring to FIG. 3, in various embodiments device **110** includes knee padding **174** coupled to housing **120**, such as to second lid **160** that is coupled to housing **120**. Knee padding **174** provides a more comfortable position for the user to rest while the user is using device **110**, such as while extending cable **136** while looking at screen **180** showing images captured by camera **140**. In various embodiments, padding **174** is formed from a foam material. In various embodiments, knee padding **174** is configured to be used by a user operating the camera **140** and display **130**.

[0037] Referring to FIG. 4, device **110** includes cover **150** detachably engaged with display **130** over an outlet, shown as USB outlet **148**. In various embodiments, cover **150** is tethered to display **130**.

[0038] Referring to FIGS. 1-5, housing **120** includes front wall **240**, an opposing back wall **242**, a first sidewall **244** and an opposing second sidewall **246**, each of the first sidewall **244** and second sidewall **246** extending between front wall **240** and back wall **242**. Device **110** includes a handle **248** coupled to housing **120**, such as pivotally coupled to housing **120**, such as pivotally coupled to front wall **240**.

[0039] Referring to FIGS. 5-6, device **110** includes bottom surface **200** defined by housing **120**, and a plurality of male couplers **202** extending from bottom surface **200** of housing **120**, the plurality of male couplers **202** detachably and rigidly couple the housing **120** to a top surface of a storage unit, such as to a storage unit in a modular system. Device **110** includes latch **230** coupled to housing **120**, such as slidably coupled to housing **120**, and latch interface **232** configured to facilitate a user actuating latch **230** upward so latch **230** does not extend below bottom surface **200**. In various embodiments, latch **230** is coupled to front wall **240** of housing **120**. In use, housing **120** is slid in a first direction until male couplers **202** engage with coupling mechanisms, such as female couplers on a top surface of a modular unit, until latch **230** engages with a latch recess defined by the modular unit. Latch **230** is configured to engage with a latch recess defined by the top surface of the storage unit the device **110** is coupled to. Engagement between the latch **230** and the latch recess biases male couplers **202** from disengaging from the female couplers, thereby result in device **110** and the modular unit being detachably and rigidly coupled together.

[0040] Referring to FIG. 6, one or more of male couplers **202** include engagement components **204** extending from and below body **206** of male coupler **202**. In various embodiments, engagement components **204** are formed from a material (e.g., rubber) that provides increased frictional resistance between housing **120** and the object that device **110** is resting on, thereby biasing device **110** from sliding with respect to the object that device **110** is resting on. In various embodiments, engagement components **204** are detachably coupled to housing **120**, and thus are replaceable by a user.

[0041] Device **110** includes bottom surface **200** defined by housing **120**. In various embodiments each of the plurality of male couplers **202** includes a body **206** extending from the bottom surface **200**, a first tongue **208**, and a second tongue **216**. The first tongue **208** and the second tongue **216** both extend from the body **206** and both are offset from, below, and parallel to the bottom surface **200**. The first tongue **208** defines a first channel **210** between the bottom surface **200** and the first tongue **208**, and the second tongue **216** defines a second channel **218** between the bottom surface **200** and the second tongue **216**. The first channel **210** and second channel **218** each extend on opposing sides (e.g., first side **224** and opposing second side **226**) of the body **206**.

[0042] Each of first channel **210** and second channel **218** includes a front open end and a back closed end. For example, first channel **210** extends from back closed end **214** to a front open end **212**, and second channel **218** extends from back closed end **222** to front open end **220**.

[0043] In various embodiments, first channel **210** extends from the closed back end **214** towards the front open end **212** and the back wall **242** of housing **120**.

[0044] In a specific embodiment, male couplers **202** and latch **230** are compatible with the coupling

WO/2017/191628, which is hereby incorporated by reference in its entirety.

[0045] Referring to FIGS. 7-8, various aspects of the internal structure of device **110** are shown. Gasket **124** engages first lid **122** and housing **120** when first lid **122** is in the closed position with respect to housing **120** to provide a seal between first lid **122** and housing **120**. In various embodiments, gasket **124** engages between first lid **122** and housing **120** to reduce the amount of debris and/or water that passes between first lid **122** and housing **120** to escape or enter first compartment **128**. Gasket **172** engages second lid **160** and housing **120** when second lid **160** is in the closed position with respect to housing **120** to provide a seal between second lid **160** and housing **120**. In various embodiments, gasket **172** engages between second lid **160** and housing **120** to reduce the amount of debris and/or water that passes between second lid **160** and housing **120** to escape or enter second compartment **164**.

[0046] First lid **122** and housing **120** collectively define first compartment **128**. First compartment **128** includes drum **142**, which secures cable **136** within first compartment **128**. In various embodiments, first lid **122** and the housing **120** collectively define a first compartment **128** configured to store a cable **136** coupled to the camera **140**.

[0047] Second lid **160** and housing **120** collectively define a compartment, such as second compartment **164**, and storage area **170**. Second lid **160** and housing **120** define second compartment, in which battery interface **166** is located. Second compartment **164** includes battery interface **166** that is configured to physically and electrically couple to rechargeable power tool battery **168**. In various embodiments, display **130** and camera **140** are configured to receive power from the rechargeable power tool battery **168** coupled to battery interface **166**. Second compartment **164** includes storage area **170** configured to store other objects and/or equipment, such as skids or other small drain cleaning accessories.

[0048] Device **110** includes arm **152** extending from housing **120** to secure cable **136** within drum **142**. In various embodiments, arm **152** extends from housing **120** towards a center of drum **142** and then downward, thereby securing cable **136** within drum **142**.

[0049] Referring to FIG. 9, in various embodiments, buttons **182** on screen **180** have a transparent or mostly transparent background and/or buttons **182** include an outline to identify the tappable area of the button **182**, such as when display **130** is a touch-screen display.

[0050] Referring to FIG. 10, in various embodiments, buttons **182** on screen **180** are brighter relative to other buttons **182** when the button **182** is selected, and buttons **182** are darker relative to other buttons **182** when the button **182** is not selected.

[0051] Referring to FIG. 11, in various embodiments, sliders **190** are large enough for a user's finger to be able to easily interact with the screen **180**. In various embodiments, one side of sliders **190** are colored and/or brighter to indicate to the user the extent that the control has been activated.

[0052] Referring to FIG. 12, in various embodiments, the screen **180** displays a status bar that indicates various pieces of important information, such as battery life remaining, Wi-Fi connectivity, and/or the time.

[0053] Referring to FIGS. 13-14, in various embodiments, screen **180** facilitates transitions for users, such as by providing a list of filter options at a top of the screen (FIG. 13), and/or by providing a pop-up alert overlaid on the screen **180**, such as semi-transparently in the middle of screen **180** (FIG. 14).

[0054] Referring to FIG. 15, in various embodiments, screen **180** includes a distance counter **186** that indicates an amount of cable **136** that has been extended out of housing **120**, and/or a zero-out button **188**.

[0055] It should be understood that the figures illustrate the exemplary embodiments in detail, and it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

[0056] Further modifications and alternative embodiments of various aspects of the disclosure will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. The construction and arrangements, shown in the various exemplary embodiments, are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. Some elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any process, logical algorithm, or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes and omissions may also be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present disclosure.

[0057] Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method claim does not actually recite an order to be followed by its steps or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is in no way intended that any particular order be inferred. In addition, as used herein, the article “a” is intended to include one or more component or element, and is not intended to be construed as meaning only one.

[0058] For purposes of this disclosure, the term “coupled” means the joining of two components directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional member being attached to one another. Such joining may be permanent in nature or alternatively may be removable or releasable in nature. As used herein, “rigidly coupled” refers to two components being coupled in a manner such that the components move together in a fixed positional relationship when acted upon by a force.

[0059] While the current application recites particular combinations of features in the claims appended hereto, various embodiments of the invention relate to any combination of any of the features described herein whether or not such combination is currently claimed, and any such combination of features may be claimed in this or future applications. Any of the features, elements, or components of any of the exemplary embodiments discussed above may be used alone or in combination with any of the features, elements, or components of any of the other embodiments discussed above.

[0060] In various exemplary embodiments, the relative dimensions, including angles, lengths and radii, as shown in the Figures are to scale. Actual measurements of the Figures will disclose relative dimensions, angles and proportions of the various exemplary embodiments. Various exemplary embodiments extend to various ranges around the absolute and relative dimensions, angles and proportions that may be determined from the Figures. Various exemplary embodiments include any combination of one or more relative dimensions or angles that may be determined from the Figures. Further, actual dimensions not expressly set out in this description can be determined by using the ratios of dimensions measured in the Figures in combination with the express dimensions set out in this description.

Claims

- 1.** A utility module comprising: a housing; a bottom surface defined by the housing; a plurality of male couplers extending from the bottom surface, the plurality of male couplers detachably and rigidly coupling the housing to a top surface of a storage unit; a display pivotally coupled to the housing; a cable configured to extend and retract from within the housing; and a camera coupled to an outer end of the cable, wherein the display displays one or more images captured by the camera.
- 2.** The utility module of claim 1, wherein each of the plurality of male couplers include a body extending from the bottom surface and a first tongue extending from the body, the first tongue offset from, below and parallel to the bottom surface.
- 3.** The utility module of claim 2, the first tongue defining a first channel comprising a front open end and a back closed end.
- 4.** The utility module of claim 3, the housing comprising a front wall and an opposing back wall, the first channel extending from the closed back end towards the front open end and the back wall, the utility module comprising a handle coupled to the front wall.
- 5.** The utility module of claim 3, the housing comprising a front wall and an opposing back wall, the first channel extending from the closed back end towards the front open end and the back wall, the utility module comprising a latch coupled to the front wall, the latch configured to engage with a latch recess defined by the top surface of the storage unit, wherein engagement the latch and the latch recess biases male couplers from disengaging from female couplers of the storage unit.
- 6.** The utility module of claim 1, wherein the display pivots with respect to the housing about a first axis and a second axis distinct from the first axis.
- 7.** The utility module of claim 6, wherein the second axis is perpendicular to the first axis.
- 8.** The utility module of claim 1, comprising a knee padding, the knee padding configured to be used by a user operating the camera and display.
- 9.** The utility module of claim 1, wherein the display is a touchscreen.
- 10.** A utility module comprising: a housing; a bottom surface defined by the housing; a plurality of male couplers extending from the bottom surface, the plurality of male couplers detachably and rigidly coupling the housing to a top surface of a storage unit; a display pivotally coupled to the housing; a camera, wherein the display displays one or more images captured by the camera; and a battery interface configured to physically couple to a rechargeable power tool battery, wherein the display and camera are configured to receive power from the rechargeable power tool battery coupled to the battery interface.
- 11.** The utility module of claim 10, comprising a first lid pivotally coupled to the housing, the first lid and the housing collectively defining a first compartment configured to store a cable coupled to the camera.
- 12.** The utility module of claim 11, comprising a second lid pivotally coupled to the housing, the second lid and the housing collectively defining a second compartment in which the battery interface is located.
- 13.** The utility module of claim 12, comprising knee padding coupled to the second lid, the knee padding configured to be used by a user operating the camera and display.
- 14.** The utility module of claim 10, wherein each of the plurality of male couplers include a body extending from the bottom surface, a first tongue extending from the body, and a second tongue opposite the first tongue and extending from the body, each of the first tongue and the second tongue offset from, below and parallel to the bottom surface.
- 15.** The utility module of claim 14, the first tongue defining a first channel and the second tongue defining a second channel, each of the first tongue and the second tongue comprising a front open end and a back closed end.
- 16.** The utility module of claim 15, the housing comprising a front wall and an opposing back wall, each of the first channel and the second channel extending from the closed back end towards the front open end and the back wall, the utility module comprising a handle coupled to the front wall.

17. The utility module of claim 15, the housing comprising a front wall and an opposing back wall, each of the first channel and the second channel extending from the closed back end towards the front open end and the back wall, the utility module comprising a latch coupled to the front wall, the latch actuating configured to engage with a latch recess defined by the top surface of the storage unit, wherein engagement the latch and the latch recess biases male couplers from disengaging from female couplers of the storage unit.

18. A utility module comprising: a housing; a bottom surface defined by the housing; a plurality of male couplers extending from the bottom surface, the plurality of male couplers detachably and rigidly coupling the housing to a top surface of a storage unit; a display pivotally coupled to the housing; a storage area defined by the housing, the storage area configured to store equipment; and a camera, wherein the display displays one or more images captured by the camera.

19. The utility module of claim 18, comprising engagement components extending from and below the plurality of male couplers, wherein the engagement components are formed from a material comprising rubber.

20. The utility module of claim 19, wherein the display pivots with respect to the housing about a first axis and a second axis distinct from the first axis, wherein the second axis is perpendicular to the first axis, and wherein the display is a touchscreen.
